Year 12 Maths - Motion with constant acceleration

Similar questions - Constant acceleration problems

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A train passes a signal P with speed 12 m s^{-1} and moves with constant acceleration along a straight horizontal track. It passes a signal Q, 400 m from P, with speed 28 m s^{-1}. It then decelerates uniformly and comes to rest at a signal R, 392 m beyond Q.

(a) Find the acceleration of the train between P and Q. **[2]**

(b) Find the time taken to travel from P to Q. **[2]**

(c) Find the deceleration between Q and R, and the time taken. **[3]**

(d) Sketch a velocity–time graph for the whole journey and find the average speed from P to R. **[4]**

2 A car starts from rest at a point O and moves along a straight road with constant acceleration 2 m s^{-2}. A motorbike passes O exactly 3 seconds after the car starts, travelling at a constant 14 m s^{-1} in the same direction.

(a) Show that the car and the motorbike are the same distance from O when $t^2 - 14t + 42 = 0$, where t is the time in seconds after the car starts. **[4]**

(b) Hence find the two times at which they are level, giving exact values. **[2]**

(c) Find the greatest distance by which the motorbike is ahead of the car, and the time at which this happens. **[4]**

Question 1

(a) $v^2 = u^2 + 2as$: $784 = 144 + 800a$, so $a = 0.8$ m s^{-2}

(b) $v = u + at$: $28 = 12 + 0.8t$, so $t = 20$ s

(c) $0 = 784 - 2d(392)$, so $d = 1$ m s^{-2} (deceleration). $0 = 28 - t$, so $t = 28$ s

(d) Graph: line rising from 12 to 28 over 20 s, then falling to 0 over 28 s. Average speed $= \dfrac{400+392}{20+28} = \dfrac{792}{48} = 16.5$ m s^{-1}

Question 2

(a) Car: $s = \frac{1}{2}(2)t^2 = t^2$. Motorbike: $s = 14(t-3)$. Equal when $t^2 = 14t - 42$, so $t^2 - 14t + 42 = 0$

(b) $t = \dfrac{14 \pm \sqrt{196-168}}{2} = 7 \pm \sqrt{7}$ (about 4.35 s and 9.65 s)

(c) Separation $= 14(t-3) - t^2 = -t^2 + 14t - 42$, greatest when $t = 7$ (midway between the two times): $-49 + 98 - 42 = 7$ m